

New Horizon College of Engineering, Bengaluru		 NEW HORIZON COLLEGE OF ENGINEERING New Horizon Knowledge Park, Ring Road, Marathalli Autonomous College Permanently Affiliated to VTU, Approved by AICTE & UGC Accredited by NAAC with 'A' Grade, Accredited by NBA	
Department: Mechanical Engineering			
Academic Semester:			
Semester: 1st	Section:	Course Code:	Course: Introduction to AI & Applications
Course Instructor:		Contact Hours /week: 04	# of credits: 03
CIE: 50		SEE: 50	Exam Hours: 3 Hrs

Prerequisites if any:			
Course Code	Course Name	Description	Semester
NA	NA		

Content delivery:	(e. g. Chalk and Talk, Power Point Presentation, Smart Board, Case Studies, Expert Lectures)
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LESSON PLAN

Lecture #	Module #	Topics	Planned Date	Actual Date	No. of Hours of Deviation (Leading/Lagging)	Reason for Deviation	Weekly Remarks by HoD with Signature
1	1	Introduction to Artificial Intelligence: Working of Artificial Intelligence					% of syllabus coverage : 10%
2		Advantages and Disadvantages of Artificial Intelligence, Evolution of Artificial Intelligence					
3		Types of Artificial Intelligence, Weak AI, Strong AI, Reactive Machines, Theory of Mind, Self-Awareness					
4		Is Artificial Intelligence Same as Augmented Intelligence and Cognitive Computing					
5		Machine Intelligence: Defining Intelligence, Components of Intelligence					% of syllabus coverage: 20%
6		Differences Between Human and Machine Intelligence					
7		Knowledge Representation and its types					
8		Experiential Learning: Case studies on Types of AI, Machine Intelligence					
9	2	Introduction to Prompt Engineering: Introduction to Prompt Engineering					% of syllabus coverage: 30%

10		The Evolution of Prompt Engineering, Types of Prompts					
11		How Does Prompt Engineering Work, Comprehending Prompt Engineering's Function in Communication					
12		The Advantages of Prompt Engineering, The Future of LLM Communication					
13		Prompts for Creative Thinking and Effective Writing: Unlocking Imagination and Innovation					% of syllabus coverage : 40%
14		Brainstorming Prompts for Problem Solving, Inspiring Prompts for Artistic Expression					
15		Igniting the Writing Process with Prompts, Prompts for Overcoming Writer's Block, Prompts for Developing Narrative Skills					
16		Experiential Learning: Activity on testing the efficiency of GPTs based on prompt mentioned					
17	3	Prompt Engineering techniques using ChatGPT and GEMINI AI and META AI: Introduction to Prompt Engineering Techniques					% of syllabus coverage : 50%
18		Instructions Prompt Technique					
19		Zero, One, and Few Shot Prompting					
20		Self-Consistency Prompt, Prompt Gallery					
21		Introduction to ChatGPT: Introduction to ChatGPT					% of syllabus coverage : 60%
22		A Concise History: From GPT-1 to GPT-4					
23		How to Install and Configure ChatGPT					
24		Experiential Learning: Activity on making students install, configure ChatGPT					
25	4	Ethics and Trends in AI: Ethical Concerns like Privacy					% of syllabus coverage : 70%
26		sustainability					
27		reliability and Bias					
28		AI as a Service (AlaaS)					
29		Recent trends in AI					% of syllabus coverage : 80%
30		Expert System, Internet of Things					
31		Artificial Intelligence of Things (AIoT)					
32		Experiential Learning: Self-study about recent trends in AI					
33	5	Industrial Applications of AI: Application of AI in Healthcare					% of syllabus coverage : 90%
34		Application of AI in Finance, Application of AI in Retail					

35		Application of AI in Agriculture, Application of AI in Education					
36		Application of AI in Transportation					
37		AI in Experimentation and Multi-disciplinary research					% of syllabus coverage:100 %
38		Robotics-an Application of AI					
39		Drones Using AI					
40		Experiential Learning: Case-study/mini-project on development of AI based physical/software model					
41		Revision Class for CIE-1					
42		Revision Class for CIE-1					
43		Revision Class for CIE-2					
44		Revision Class for CIE-2					
45		Revision Class for SEE					
46		Revision Class for SEE					
47		Revision Class for SEE					
48		Revision Class for SEE					
49		Revision Class for SEE					
50		Revision Class for SEE					
51		Revision Class for SEE					
52		Revision Class for SEE					

Innovative Pedagogies

Module #	Innovative Teaching Activities	Planned Date	Actual Date
1	Case studies on Types of AI, Machine Intelligence		
2	Activity on testing the efficiency of GPTs based on prompt mentioned		
3	Activity on making students install, configure Chat-GPT		
4	Case-study/mini-project on development of AI based physical/software model		

Alternate Assessment Techniques

Alternate Assessment Techniques Description	Planned Date	Actual Date
AAT1		
AAT2		

Assessment and Evaluation

Continuous Internal Evaluation (50 Marks)										
COs	IA1		IA2		AAT1-A		AAT1-B (If Applicable)		AAT2	
	RBT	MARKS	RBT	MARKS	RBT	MARKS	RBT	MARKS	RBT	MARKS
C01	L1	20			L2	2.5			L3	2
C02	L2	20			L3	2.5			L4	2
C03	L4	10	L2	10	L4	2.5			L4	2
C04			L3	20			L2	2.5	L3	2
C05			L4	10			L3	2.5	L3	1
C06			L2	10			L4	2.5	L4	1
Total	50		50		15				10	

Semester End Examination (50 Marks)										
COs	Module 1		Module 2		Module 3		Module 4		Module 5	
	RBT	MARKS	RBT	MARKS	RBT	MARKS	RBT	MARKS	RBT	MARKS
C01	L1	10								
C02			L2	10						
C03					L3	10				
C04							L2,L4	10		
C05									L2	05
C06									L4	05
Total		10		10		10		10		10

Faculty

HoD